First Two-Class Appointment Simulation: Day-Level Formulation v2

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1 Purpose and Reader Map

This v2 note is the day-level replacement for the earlier slot-level formulation. The earlier note remains useful because it showed the model through concrete pictures: a time-evolution graph, a rolling calendar table, a cancellation-hazard plot, and a worked FCFS appointment book. This version keeps that teaching style, but changes the model state to day/class counts.

Core Modeling Choice

The simulation clock is the day D. The model state records how many patients of each class are scheduled on each residual day. The code may keep internal slot positions to enforce capacity and audit utilization, but slot position m is no longer a model state variable.

1.1 What Changed from the Previous Note

The table below records what the old note used to help a reader understand the simulator and the equivalent object in the day-level formulation.

Old explanatory device	What it taught	Day-level v2 replacement
Slot epoch $t = (D, s)$ and rollover graph	how the simulator moved through slots inside a day	daily epoch graph $D \to D + 1$ with the seven explicit daily events
Slot calendar $Y_t(r, m)$	where each booked appointment sat in the rolling calendar	class-count state $X^D_{i,r}$ plus aggregate occupancy N^D_r by residual day
Colored $H \times S$ appointment book	how FCFS filled slots and how holes appeared	colored residual-day table showing total load, open capacity, and future-only offers
No-show and cancellation plots	how baseline step rules differ from the smoother legacy delay-sensitive functions	baseline $\xi_i(\tau)$ and ϕ_i in solid lines, with advanced $\tilde{\xi}_i(\tau)$ and $\tilde{\phi}_i(\tau, r)$ shown as dashed later options
Worked FCFS example	how a request becomes an offer, booking, balk, no-show, or service	day-level progression example with cancellations, arrivals, offers, balking, service, metrics, and calendar advance
Metric tables	how logs map to access and utilization measures	same metrics, plus the `daily_progression` audit table that records every daily step

2 Model at a Glance

The simulator keeps the same operational feedback loop as the earlier model and the motivating appointment-queueing literature.

Core Feedback Loop

$$\begin{aligned} \text{more backlog} &\Rightarrow \text{larger offered delay} \\ &\Rightarrow \text{more balking, cancellation, or no-show risk} \\ &\Rightarrow \text{lower realized service.} \end{aligned}$$

The day-level objects are:

- D: current calendar day.
- S: number of appointment slots available per day.
- H: rolling booking horizon in days.
- $\mathcal{I}$: set of patient classes, equal to $\{1, 2\}$ in the first experiments.
- $r \in \{0, 1, \ldots, H-1\}$: residual delay, where $r = 0$ means appointments scheduled for today.
- τ: original offered delay accepted by a booked patient.

The main distinction remains:

Residual Delay vs. Original Delay

Residual delay r belongs to the day-level state and decreases as days pass. Original delay τ is stored with the booked patient because balking, optional advanced cancellation rules, and no-show behavior may depend on the delay the patient accepted.

3 Day-Level State

At the start of day D, after the daily cancellation step and before processing new arrivals, define

$$X_{i,r}^D = \text{number of class-}i\text{ patients scheduled for day } D + r.$$

The state vector is

$$X^D = \left(X_{i,r}^D : i \in \mathcal{I},\ r = 0, \ldots, H-1\right).$$

For each residual day, aggregate occupancy is

$$N_r^D = \sum_{i \in \mathcal{I}} X_{i,r}^D,$$

and daily capacity imposes

$$N_r^D \leq S.$$

Slot positions inside a day are not part of X^D. The implementation still keeps an internal list of S slots per day so it can find open capacity, produce service-slot logs, and compute empty-slot utilization. That internal detail is an audit layer, not the mathematical state.

3.1 Patient Audit Record

When a class-i patient accepts an offer with delay τ, the state count is incremented on the selected residual day, and the implementation also stores a patient audit record

$$(i, \tau).$$

The audit record is needed because:

- $b_i(\tau)$ determines balking when the offer is made;
- baseline $\xi_i(\tau)$ determines no-show at appointment time;
- advanced no-show experiments may instead use $\tilde{\xi}_i(\tau)$;
- optional advanced cancellation functions may use $\tilde{\phi}_i(\tau, r)$;
- cohort logs and delay metrics need the originally offered delay.

4 Patient Classes and Behavior

Each class is described by

$$C_i = \{\lambda_i, b_i(\tau), \phi_i, \xi_i(\tau)\}.$$

Object	**Meaning**
λ_i	expected number of class-i arrivals per day
$b_i(\tau)$	probability that class-i declines an offered appointment with delay τ
ϕ_i	scalar daily cancellation probability for class-i scheduled patients
$\tilde{\phi}_i(\tau, r)$	advanced daily cancellation rule using original delay and residual delay; not used in the baseline FCFS analysis
$\xi_i(\tau)$	baseline step no-show probability at appointment time if the patient has not canceled
$\tilde{\xi}_i(\tau)$	advanced delay-sensitive no-show rule; not used in the baseline FCFS analysis

The baseline cancellation input is ϕ_i. It is applied directly once per day to every scheduled class-i patient considered in the daily cancellation step. It is not converted from an eventual cancellation probability.

For two-class experiments, demand can also be parameterized by total daily arrival volume λ and class-1 share p:

$$\lambda_1 = p\lambda, \qquad \lambda_2 = (1-p)\lambda.$$

4.1 Working Behavior Profile

The first FCFS experiments use a concrete two-class profile. The numbers below are not required by the model, but they make the figures and examples reproducible.

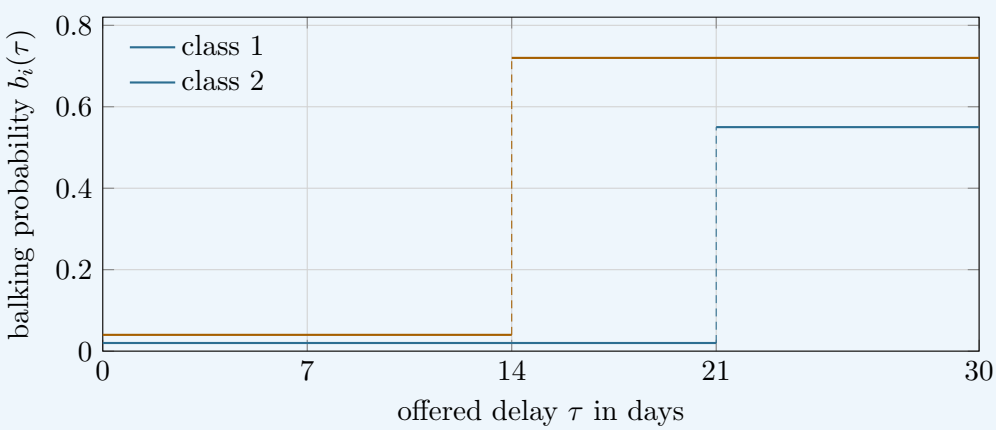
Object	Class 1: MRI-like diagnostic	Class 2: behavioral-health follow-up
Daily demand	$\lambda_1 = 3.5$ when $\lambda = 6.0$ and $p = 7/12$	$\lambda_2 = 2.5$ when $\lambda = 6.0$ and $p = 7/12$
Balking	low until a 21-day access threshold, then high	low until a 14-day access threshold, then high
No-show	step $\xi_1(\tau)$: 0.01 before 21 days, 0.31 from 21 days onward; dashed $\tilde{\xi}_1(\tau)$ reserved for later	step $\xi_2(\tau)$: 0.15 before 14 days, 0.51 from 14 days onward; dashed $\tilde{\xi}_2(\tau)$ reserved for later
Cancellation	scalar $\phi_1 = 0.01$ in the baseline; advanced $\tilde{\phi}_1(\tau, r)$ is reserved for later experiments	scalar $\phi_2 = 0.02$ in the baseline; advanced $\tilde{\phi}_2(\tau, r)$ is reserved for later experiments

4.2 Balking Function

In the current FCFS baseline, balking is represented with a step function of the offered delay. This is intentionally simple: patients are fairly willing to accept appointments below a practical access threshold, and much less willing once the offered delay crosses that threshold.

$$b_1(\tau) = \begin{cases} 0.02, & \tau < 21, \\ 0.55, & \tau \geq 21, \end{cases} \qquad b_2(\tau) = \begin{cases} 0.04, & \tau < 14, \\ 0.72, & \tau \geq 14. \end{cases}$$

Balking as a Step Function of Delay



The operational interpretation is direct. Under FCFS, congestion increases the first available future residual day. Once that offered delay crosses a class-specific threshold, many more patients decline the offer. This is the access-side version of the same delay feedback loop that the literature usually models through no-show behavior.

4.3 No-Show Function

No-show behavior uses the original accepted delay τ, not the residual delay r on a later day. In the current FCFS baseline, no-show is represented with a step function, just like balking:

$$\xi_1(\tau) = \begin{cases} 0.01, & \tau < 21, \\ 0.31, & \tau \geq 21, \end{cases} \qquad \xi_2(\tau) = \begin{cases} 0.15, & \tau < 14, \\ 0.51, & \tau \geq 14. \end{cases}$$

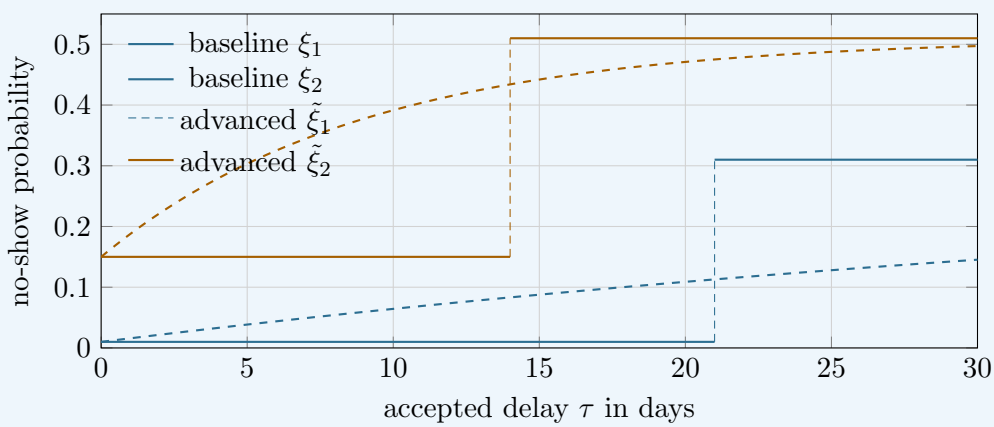
The earlier Green-Savin style delay-sensitive curve is still useful as an advanced option for later experiments. In this note it is written as

$$\tilde{\xi}_i(\tau) = \gamma_{i,\max} - (\gamma_{i,\max} - \gamma_{i,0})e^{-\tau/s_i}.$$

For the same illustrative values,

$$(\gamma_{1,0}, \gamma_{1,\max}, s_1) = (0.01, 0.31, 50), \qquad (\gamma_{2,0}, \gamma_{2,\max}, s_2) = (0.15, 0.51, 9).$$

Baseline Step No-Show and Advanced $\tilde{\xi}$



The solid curves are the implemented baseline. They make the daily progression easier to read: patients who accepted a short delay behave one way, and patients who accepted a longer delay behave another way. The dashed curves show the smoother advanced $\tilde{\xi}_i(\tau)$ option, which is closer to Green-Savin style delay sensitivity and remains available for later experiments.

Cancellation Probability Figure: Baseline and Advanced Rule

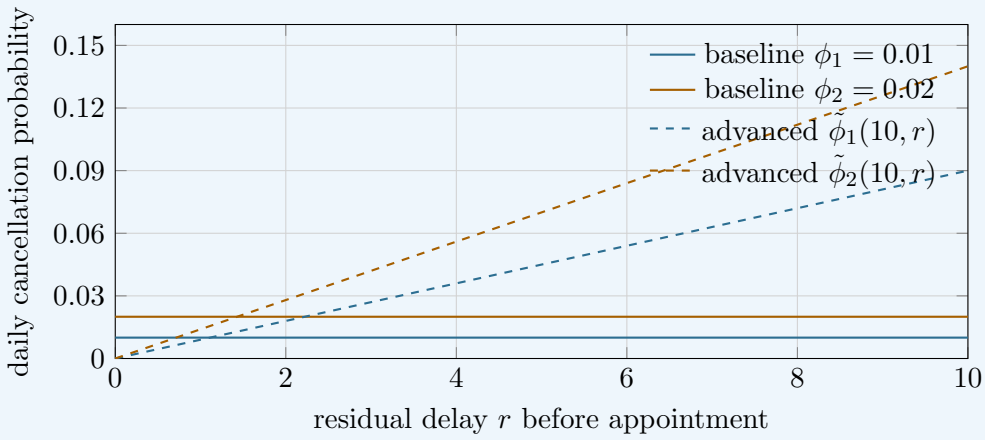
The new baseline uses a constant daily cancellation probability ϕ_i. The older delay/residual-delay function is renamed

$$\tilde{\phi}_i(\tau, r),$$

and is kept only as an advanced option for later experiments. It is not part of the baseline FCFS analysis in this note.
For example, the advanced taper has the form

$$\tilde{\phi}_i(\tau, r) = \min\{a_i + c_i\tau, \bar{\phi}_i\}\frac{r}{\tau}, \qquad 1 \leq r \leq \tau,$$

with $\tilde{\phi}_i(\tau, 0) = 0$. The figure shows the difference between the scalar baseline and that advanced option.

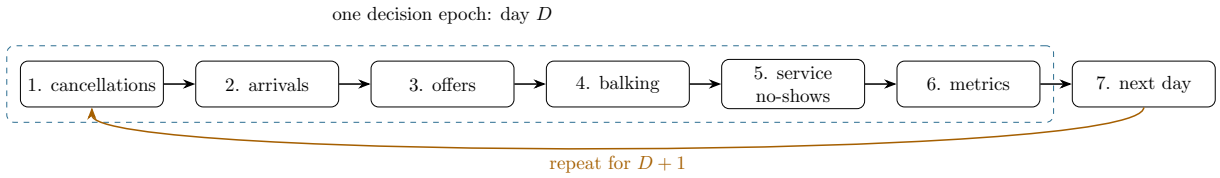


The flat lines are the implemented baseline: each scheduled class-i patient faces the same daily probability ϕ_i on each day they are considered. The dashed curves are only an example of the advanced $\tilde{\phi}_i(\tau, r)$ rule, where risk can vary with both the original delay and the residual delay.

5 Daily Simulation Order

The simulator advances one day at a time. For each day D, it executes the following steps in this exact order.

1. **Compute cancellations for the day.** Each already scheduled class-i patient independently cancels with daily probability ϕ_i.
2. **Simulate daily arrivals.**
$$M_i^D \sim \text{Poisson}(\lambda_i),$$
where M_i^D is the number of class-i arrivals on day D.
3. **Offer future days in FCFS order.** Arrivals are processed sequentially in randomized within-day order. The baseline policy offers the earliest future day $D + r$, $r \in \{1, \ldots, H - 1\}$, with open capacity.
4. **Simulate balking.** A class-i patient offered delay $\tau = r$ accepts with probability $1 - b_i(\tau)$. If the patient balks, the capacity remains open for later arrivals.
5. **Simulate no-shows and service for day D.** Each remaining appointment scheduled for D no-shows with baseline probability $\xi_i(\tau)$, using its stored original delay.
6. **Compute metrics for day D.** The simulator records access, booking, cancellation, service, empty-slot, utilization, and delay metrics.
7. **Advance to day $D + 1$.** The current day is removed, residual delays decrease by one, and a new empty future day is appended.



Future-Only Offers

New arrivals on day D cannot book day D. Even if a morning cancellation opens capacity today, baseline FCFS searches only $r = 1, \ldots, H-1$. Same-day holes therefore affect today's empty-slot metrics, not today's new offers.

6 Day-Level Calendar Picture

The old note used an $H \times S$ slot calendar to show exactly where patients sat. The day-level model uses the same rolling calendar concept, but it displays counts by class and residual day.

Visualizing the Day-Level State X^D

Suppose $S = 3$, $H = 4$, and after the cancellation step on day D the class-count state is:

Count	**day** D $r = 0$	**day** $D+1$ $r = 1$	**day** $D+2$ $r = 2$	**day** $D+3$ $r = 3$
$X^D_{1,r}$	1	2	0	0
$X^D_{2,r}$	1	1	1	0
N^D_r	2	3	1	0
$S - N^D_r$	1	0	2	3

The current day $r = 0$ has one open slot, but it is not bookable by new arrivals. Tomorrow ($r = 1$) is full. The first eligible future day with capacity is $r = 2$, so the first offer under FCFS is day $D + 2$, with $\tau = 2$.

7 FCFS Booking Rule

For a new arrival on day D, the baseline policy searches

$$r = 1, 2, \ldots, H-1$$

and offers the first residual day satisfying

$$N^D_r < S.$$

If no such day exists, the patient receives no offer. If the patient accepts, the simulator increments $X_{i,r}^D$ and stores the patient's original delay as $\tau = r$. If the patient balks, $X_{i,r}^D$ is not changed and the capacity remains available to later arrivals.

8 Worked Day-Level Example

This example is the day-level counterpart of the previous note's FCFS appointment-book example. It shows the state, the patient audit record, the booking decision, the no-show/service step, and the next-day rollover.

Worked Example: One Full Day D

Take $S = 3$, $H = 4$, and the post-cancellation state from the previous table:

$$N_0^D = 2, \qquad N_1^D = 3, \qquad N_2^D = 1, \qquad N_3^D = 0.$$

Assume that the cancellation step has already removed one appointment, so the daily cancellation counter is 1.
Daily arrivals are drawn once per class:

$$M_1^D = 2, \qquad M_2^D = 1.$$

The simulator randomizes the within-day order. Suppose the realized order is:

$$\text{class } 1, \quad \text{class } 2, \quad \text{class } 1.$$

The three offer decisions are:

Arrival	Class	FCFS offer	Result
1	1	$r = 2,\ \tau = 2$	accepts; $X_{1,2}^D$ increases by 1
2	2	$r = 2,\ \tau = 2$	balks; the day $D + 2$ capacity remains available
3	1	$r = 2,\ \tau = 2$	accepts; $X_{1,2}^D$ increases by 1

After offers and balking, the aggregate future counts are:

$$N_0^D = 2, \qquad N_1^D = 3, \qquad N_2^D = 3, \qquad N_3^D = 0.$$

The two accepted patients are stored in the audit layer as $(1, 2)$ and $(1, 2)$. The balked class-2 patient creates no appointment record.
Now the simulator resolves today's $r = 0$ appointments. Suppose one scheduled patient attends and one no-shows. Since $S = 3$ and only two patients were scheduled for today, the daily service outcome is:

$$\text{served} = 1, \qquad \text{no_shows} = 1, \qquad \text{empty_slots} = 1.$$

Then the calendar advances. The full day $D + 1$ becomes the new current day, residual delays decrease by one, and a new empty day $D + 4$ is appended:

$$\left(N_0^{D+1}, N_1^{D+1}, N_2^{D+1}, N_3^{D+1}\right) = (3, 3, 0, 0).$$

Same Example as a Daily Progression Audit

The implementation exposes the same sequence through `daily_progression`. A stylized row for each step of the example is:

Step	Arr	Offer	Book	Balk	Canc	Served	NS	Empty	Interpretation
cancellations	0	0	0	0	1	0	0	0	one appointment removed before arrivals
arrivals	3	0	0	0	1	0	0	0	daily Poisson counts have been drawn
offers and balking	3	3	2	1	1	0	0	0	all arrivals got an offer; one declined
no-shows/service	3	3	2	1	1	1	1	1	today's scheduled patients are resolved
metrics	3	3	2	1	1	1	1	1	daily counters are ready for summaries
advance	3	3	2	1	1	1	1	1	old $r = 1$ becomes the new $r = 0$

This table is not a second model. It is an audit view of the model order. It exists so a reader can verify that the simulator is progressing exactly one day at a time.

9 Performance Measures

For patients arriving during the measured window, the simulator reports:

Metric	Definition
Arrivals	number of appointment requests
Booked	number of requests that accept an offer
Balked	number of requests that receive an offer and decline it
No offer	number of requests for which no future capacity is available
Canceled	number of booked patients who cancel before service resolution
No-shows	number of booked patients who remain scheduled but do not attend
Served	number of booked patients who attend
Mean booked delay	average original offered delay τ among booked patients

The main access and service ratios are:

$$\frac{\text{Booked}}{\text{Arrivals}}, \qquad \frac{\text{Served}}{\text{Booked}}, \qquad \frac{\text{Booked within target}}{\text{Arrivals}}.$$

For measured service days, daily capacity is still S. Slot utilization is therefore:

$$\rho^{\text{booked}} = \frac{\text{booked slots}}{S \times \text{measured days}}, \qquad \rho^{\text{attended}} = \frac{\text{served slots}}{S \times \text{measured days}}.$$

10 Implementation Mapping

The code follows this day-level formulation:

- `arrival_rate` is λ_i, the expected class-i arrivals per day.
- `cancel_probability` as a scalar is the direct daily cancellation probability ϕ_i.
- `tau_offered` and `tau_booked` are original offered delays in days.
- `arrival_slot` is retained as a legacy log column and is empty under the day-level process.
- `state_log` records $X^D_{i,r}$ for every configured class and residual day.
- `daily_progression` records the seven daily model steps for each measured day.
- `slots_per_day` is daily capacity S, and remains the denominator for slot utilization.

Review Against the Intended New Model

- The model state is day/class count state $X^D_{i,r}$, not slot state $Y_t(r, m)$.
- The arrival rate λ_i is daily, so $M^D_i \sim \text{Poisson}(\lambda_i)$.
- Same-day offers are disabled; bookable residual days are $r = 1, \ldots, H-1$.
- Scalar ϕ_i is a daily cancellation probability applied once per day.
- A booked patient's original τ is retained for balking, no-show, advanced cancellation rules, and delay metrics.
- The event order is cancellations, arrivals, offers, balking, no-shows/service, metrics, and calendar advance.

11 Relation to the Literature

Green and Savin (2008), Liu and Ziya (2014), and Liu (2015) use stylized queue, backlog, or window states to study delay-sensitive attendance and appointment access. The day-level simulator is a calendar version of the same mechanism. The state $X^D_{i,r}$ records the future schedule by class and residual day; the offered delay τ is the calendar analog of the congestion seen by the patient at booking.

11.1 Common Mechanism

The analytical papers usually compress the appointment book into one congestion variable, such as queue length, backlog, or the number of appointments already ahead of a new patient. This note keeps the appointment book explicit but day-level:

$$X^D = \left(X^D_{i,r} : i \in \mathcal{I},\ r = 0, \ldots, H-1\right).$$

Under FCFS, this state implies the offered delay

$$\tau = \min\{r \in \{1, \ldots, H-1\} : N^D_r < S\},$$

when such an r exists. Thus τ is the bridge between the calendar model and the queueing literature. A larger backlog pushes the first feasible r farther into the future, and that larger τ then enters $b_i(\tau)$ and the baseline step no-show rule $\xi_i(\tau)$. In later experiments, the smoother advanced rule $\tilde{\xi}_i(\tau)$ can be used instead. In the baseline, cancellation is the scalar daily probability ϕ_i. Later experiments can replace it with the advanced rule $\tilde{\phi}_i(\tau, r)$.

Why the Day-Level Model Still Matches the Literature

The literature does not require a slot index to create delay feedback. It requires a congestion state that determines the delay seen at booking. In v2, X^D is that state, and FCFS converts it into the offered delay τ.

11.2 Green and Savin (2008)

Green and Savin study an appointment queue in which no-show behavior worsens with the backlog seen when the appointment is made. Their no-show curve is an exponential-saturation function of a congestion measure. The v2 baseline deliberately simplifies that behavior to a step function $\xi_i(\tau)$ so the daily progression stays easy to read, but it keeps the Green-Savin shape as the advanced option

$$\tilde{\xi}_i(\tau) = \gamma_{i,\max} - (\gamma_{i,\max} - \gamma_{i,0})e^{-\tau/s_i}.$$

The mapping is:

- their backlog variable maps to the FCFS delay implied by X^D;
- their no-show mechanism maps directly to $\tilde{\xi}_i(\tau)$, while the baseline $\xi_i(\tau)$ is a step approximation used for the first FCFS experiments;
- their homogeneous queue is extended here to multiple classes through i;
- v2 also adds explicit balking and pre-appointment cancellation.

The simplification in v2 is that time advances by day rather than by individual service slot. This is acceptable for the first simulator because the central feedback loop depends on appointment delay in days, not on the exact within-day slot where the appointment sits.

11.3 Liu and Ziya (2014)

Liu and Ziya analyze panel size and overbooking decisions under patient no-shows. Their state is more stylized than a rolling calendar, but the core tradeoff is the same: more demand can increase utilization, while longer waits can reduce realized attendance.

The v2 simulator intentionally does not optimize panel size or overbooking yet. It fixes daily capacity S, uses FCFS, and observes how access, booking, cancellation, no-show, and service metrics respond to demand and behavior. This makes the simulator a diagnostic baseline before any future optimization layer is added.

The mapping is:

- their panel size / demand pressure maps to total daily arrival rate λ;
- their show-up probability maps to $1 - \xi_i(\tau)$ in the baseline and $1 - \tilde{\xi}_i(\tau)$ in the more detailed variant;
- their capacity decision maps here to fixed daily capacity S;
- their overbooking decision is not active in the v2 FCFS baseline.

11.4 Liu (2015)

Liu studies the optimal appointment scheduling window under delay-dependent no-shows. The model's window size limits how far into the future appointments may be scheduled. In v2, the

rolling horizon H plays the same operational role:

$$r \in \{0, 1, \ldots, H-1\}.$$

Because same-day offers are disabled in the current baseline, the bookable window is

$$r \in \{1, \ldots, H-1\}.$$

If every future residual day is full, the arriving patient receives no offer. This is the explicit calendar equivalent of a finite scheduling window rejecting or deferring demand when the window is full.

The mapping is:

- Liu's appointment window K maps to the simulator horizon H;
- delay-sensitive attendance maps to $\xi_i(\tau)$ in the baseline and to $\tilde{\xi}_i(\tau)$ in the more detailed variant;
- the no-offer event appears when the FCFS search over $r = 1, \ldots, H-1$ finds no capacity;
- optimizing H is a later experiment, not part of the current FCFS-only baseline.

11.5 Why Only FCFS Here

The old codebase already supports alternative slot-selection policies, but this note deliberately keeps the analysis focused on FCFS. That is the cleanest first experiment because it isolates the baseline delay-feedback mechanism:

$$X^D \to \tau_{\text{FCFS}} \to b_i(\tau),\ \phi_i,\ \xi_i(\tau) \to \text{daily outcomes}.$$

Future policies can be introduced later as changes to the arrival, admission, or offer step. They should be compared against this FCFS baseline only after the day-level mechanics and metrics are stable.

Model / paper	**State abstraction**	**Delay variable**	**Difference from this simulator**
Green and Savin (2008)	queue length / backlog	backlog seen when appointment is made	analytical single-class queue; no explicit calendar
Liu and Ziya (2014)	queue length in appointment model	congestion at booking time	optimizes panel size and overbooking
Liu (2015)	finite appointment window	delay implied by the scheduling window	optimizes the window size K
This simulator v2	rolling day-count calendar $X^D_{i,r}$	original offered delay τ	two classes, cancellations, balking, and explicit daily progression

12 References

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